specifying application code
adding new features 273–274
describing expected behavior as failing test 271
making test pass 271–272
refactoring 272
Specs2 291–293
Spock 50, 291–293
Spring 251
stakeholders 80–82
state, maintaining between step definitions 149–150
step definitions
for Behave 167–168
for Cucumber-JS 176–177
for Cucumber-JVM 163
example-based scenarios 151–152
exploring low-level requirements 275–277
implementing features 268
for JBehave 156–157
keeping methods simple 145–146
maintaining state 149–150
overview 143–145, 147–148
passing parameters to 148–149
scenario outcomes 152–153
for SpecFlow 172–173
using table data from 150–151
step library 197
@Steps annotation 197
stories
breaking features down into 37
defined 33
illustrating with examples 38
stubs 281–283
Surefire 332
system design specification. *See* SDS

T

tables of examples
for Behave 169
for Cucumber-JVM 165
for JBehave 158–159
low-level requirements using 277–279
overview 125–128
for SpecFlow 173–174
tabular parameters 158
tags
annotating scenarios with 137–139
organizing living documentation using 311–312
task boards 308
TDD (Test-Driven Development)
BDD and 12–14
BDD built on practices of 264
overview 261–263
TeamCity 335
Technical layer 198–199, 223
technical living documentation
for legacy applications 319–320
overview 315–316
unit tests as 316–318
@Test annotation 288
test suite 183
Test-Driven Development. *See* TDD
testing
getters/setters 272
parallel
within build 332–335
on multiple machines 335–337
using Selenium Grid 337–343
See also automated acceptance tests
testing UI
with headless browsers 204–205
how much testing to do 205–206
page objects
exposing simple types and domain objects 227–230
fluent selectors and 234–235
libraries for 233–234
navigating with 232
overview 222–223
Page Objects pattern 223–224
reporting on page state 230–232
WebDriver API and 224–227
Selenium WebDriver
AJAX applications 219–221
element interaction methods 218–219
identifying elements 210–218
overview 206–207
test-friendly web applications 221–222
using Java API 207–210
too many tests and 203–204
The Grinder 256
@Then annotation 47, 156
then keyword 39
Then step 122, 131
[Then] attribute 172
Thucydides 120, 136, 154, 157, 233
tools, unit testing
Jasmine 289–290
NSpec 290–291
RSpec 287–289
Specs2 291–293
Spock 291–293
traditional 285–287
train timetable application
determining project objectives 35
implementing features
automated tests 46–55
executable specifications 42–46
going from examples to acceptance criteria 39–40
setting up Maven and Git 40–41
tests as living documentation 55–56
maintenance 56–58
overview 33–34
requirements analysis
breaking features down into stories 37
describing features 35–37
illustrating stories with examples 38
@transfers annotation 169
Transport for NSW 73
Twill 204

U

UAT (User Acceptance Testing) 322, 329
ubiquitous 268